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United States Patent	12397358
Kind Code	B2
Date of Patent	August 26, 2025
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System and method for deep drilling into a pet panel and pet panel including a deep drilled hole

Abstract

A method for deep drilling into a panel composed of compressed polyester fibers includes providing a guide system, aligning the guide system with the panel, applying a force to a face of the panel, and drilling a hole into the edge of the panel with the guide system and the drill while the force is applied to the face of the panel. The guide system supports a drill having a drill bit for movement in an axial direction. The force is applied to an area of the panel that is in axial alignment with the guide system.

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Appl. No.: **17/734805**

Filed: **May 02, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20230347420 A1	Nov. 02, 2023

Publication Classification

Int. Cl.: **B23B49/00** (20060101); **B23B41/00** (20060101); **B23B41/02** (20060101)

U.S. Cl.:

CPC **B23B49/00** (20130101); **B23B41/00** (20130101); **B23B41/02** (20130101);

Field of Classification Search

CPC: B23B (41/02); B23B (45/003); B23B (47/28); B23B (47/287); B23B (49/02); B23B (2215/64); B23B (2226/27); B23B (2226/61); B23B (35/00); B23B (51/04)

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Background/Summary

FIELD
(1) The present disclosure relates to PET panels and to methods for deep drilling into PET panels.
BACKGROUND
(2) Panels are used in office settings to separate or divide between individual employee workspaces. In some scenarios, multiple panels may be used to enclose the workspaces as desired. Such panels may require means of assembling to enclose the workspaces. Current panels may use brackets or other attachments on outsides of the panels to assemble together. The attachments are placed on the outside of the panels because consistent and precise hole drilling into relatively thin panels has a low success rate. For example, current drilling methods typically result in holes having undesired extension directions due to drill bit wobble and/or insertion error, among other sources of error.
SUMMARY
(3) In one aspect, the disclosure provides a method for deep drilling into a panel composed of compressed polyester fibers. The panel has an edge and a face. The method includes providing a guide system. The guide system supports a drill having a drill bit for movement in an axial direction. The method also includes aligning the guide system with the panel and applying a force to the face of the panel. The force is applied to an area of the panel that is in axial alignment with

the guide system. The method further includes drilling a hole into the edge of the panel with the guide system and the drill while the force is applied to the face of the panel.
(4) In another aspect, the disclosure provides a drilling kit for deep drilling into a panel composed of compressed polyester fibers. The panel has a face and an edge. The drilling kit includes a drill, a drill bit coupled to the drill, and a guide system. The guide system includes a support rail and a guide block configured to be positioned adjacent the edge of the panel. The guide block receives at least a portion of the drill bit to guide movement of the drill bit relative to the panel. The guide system also includes a slider supporting the drill. The slider is movable in an axial direction along the support rail to direct the drill toward and away from the panel. The drilling kit also includes a weight configured to apply a force to the face of the panel. The force is configured to be applied to an area of the panel that is in axial alignment with the guide system.
(5) In another aspect, the disclosure provides a panel including a body composed of compressed polyester fibers. The body has a first face, a second face opposite the first face, a first edge extending between the first face and the second face, and a second edge extending between the first face and the second face. The second edge is opposite the first edge. The body also has a hole drilled through the first edge and extending at least partially toward the second edge. The hole has a diameter and a length. The length is at least 10 times the diameter.
(6) In another aspect, the disclosure provides a drill bit for deep drilling a hole in a panel composed of compressed polyester fibers. The drill bit includes a generally hollow body. The generally hollow body includes a smooth outer surface, a first end configured to be coupled to a drill, and a second end opposite the first end. The second end has a sharpened edge without cutting teeth. The generally hollow body also includes an aperture extending from the second end toward the first end for receiving material removed from the panel during a drilling operation. The generally hollow body has a length that is at least 10 times a diameter of the generally hollow body.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view of a screen including a panel.
(2) FIG. **2** is a perspective, cross-sectional view of the panel of FIG. **1**.
(3) FIG. **3** is a front, cross-sectional view of the panel of FIG. **1**.
(4) FIG. **4** is a bottom view of the panel of FIG. **1**.
(5) FIG. **5** is a zoomed in view of a structural insert for the panel of FIG. **1**.
(6) FIG. **6** is a front view of a plurality of structural inserts for the panel of FIG. **1**.
(7) FIG. **7** is a perspective view of a plurality of workspaces including screens according to another embodiment of the disclosure.
(8) FIG. **8** is a perspective view of a workspace including a screen according to yet another embodiment of the disclosure.
(9) FIG. **9** is a perspective view of a workspace including a screen according to even yet another embodiment of the disclosure.
(10) FIG. **10** is a perspective view of a drilling kit for deep drilling a hole into a panel according to any of the embodiments of the disclosure.
(11) FIG. **11** is an enlarged perspective view of a portion of the drilling kit of FIG. **10**.
(12) FIG. **12** is a perspective view of a drill bit for use with the drilling kit of FIG. **10**.
(13) FIG. **13** is an enlarged view of a portion of the drill bit of FIG. **12**.
(14) FIG. **14** is a flow chart of a method for deep drilling into a panel.
(15) FIG. **15** is a perspective view of providing and aligning a guide system with the panel according to the method of FIG. **14**.
(16) FIG. **16** is a perspective view of applying a force to the panel according to the method of FIG.

(17) FIG. **17** is a perspective view of a first progression point of drilling a hole into the panel according to the method of FIG. **14**.
(18) FIG. **18** is a perspective view of a second progression point of drilling the hole into the panel according to the method of FIG. **14**.
DETAILED DESCRIPTION
(19) Before any embodiments of the disclosure are explained in detail, it is to be understood that the invention is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The invention is capable of other embodiments and of being practiced or of being carried out in various ways. In addition, terms such as “approximately,” “about,” and the like associated with values should be understood to encompass rounding and/or manufacturing tolerances associates with the values.
(20) FIG. **1** illustrates a screen **14**. The illustrated screen **14** includes one or more panels **10**. The panels **10** that form the screen **14** are composed of compressed polyester fibers (e.g., compressed polyester staple fibers). As such, the panel **10** is a PET, or polyethylene terephthalate, panel. Although PET is normally a clear plastic, the panel **10** may be composed of compressed PET fibers, making the panel **10** opaque. Additionally, PET is a relatively strong and lightweight plastic. In at least one application, the screen **14** may be used in an office. For example, the screen **14** may be used as a divider for an employee workspace **16** within an office. The screen **14** is operable to block the line of sight into and out of the workspace **16**. The screen **14** may also or alternatively be used as a privacy screen, a modesty screen, and/or a room divider that is part of the workspace **16** or that is a standalone screen. As such, the screen **14** may advantageously increase the privacy and reduce distractions for an employee utilizing the workspace **16**.
(21) As illustrated in FIG. **2**, the panel **10** includes a body **18**. The illustrated body **18** is composed of compressed polyester fibers such that the panel **10** is a PET panel. The body **18** includes a first face **20** (FIG. **4**), a second face **22**, a first edge **26**, a second edge **30**, a third edge **32**, and a fourth edge **34**. The first face **20** is positioned opposite the second face **22**. The first edge **26** extends between the first face **20** and the second face **22**. The second edge **30** extends between the first face **20** and the second face **22**. The second edge **30** is positioned opposite the first edge **26**. The first edge **26** extends parallel to the second edge **30**. The third edge **32** and the fourth edge **34** also extend between the first face **20** and the second face **22**. Each of the third edge **32** and the fourth edge **34** extends transverse to both the first edge **26** and the second edge **30**. The fourth edge **34** is positioned opposite the third edge **32**. The fourth edge **34** extends parallel to the third edge **32**. In the illustrated embodiment, body **18** of the panel **10** forms a generally rectangular prism. In other embodiments, the body **18** of the panel **10** may form a panel **10** that has a different shape such as, but not limited to, a cylinder, a triangular prism, a semi-circular prism, a trapezoidal prism, a square prism, or the like. In such embodiments, the panel **10** may include fewer or more edges than the body **18** of the illustrated embodiment. For example, in the embodiment in which the body **18** forms a semi-circular prism, the panel includes an edge and a curved surface that extends from each end of the edge such that the body only includes two edges.
(22) With reference to FIGS. **3** and **4**, the body **18** of the panel **10** has a body length L.sub.B and a body thickness T.sub.B. The distance between the first edge **26** and the second edge **30** defines the body length L.sub.B. In some embodiments, the body length L.sub.B may be at least 1 foot. In other embodiments, the body length L.sub.B may be at least 3 feet. In still other embodiments, the body length L.sub.B may be more than 3 feet. The distance between the first face **20** and the second face **22** defines the body thickness T.sub.B. In some embodiments, the body thickness T.sub.B may be at least 0.5 inches. In other embodiments, the body thickness T.sub.B may be less than 1 inch. In the illustrated embodiment, the body thickness T.sub.P may be between 0.5 inches and 1 inch. In other embodiments, the body thickness T.sub.B of the body may be less than 0.5

inches. In further embodiments, the body thickness T.sub.B may be greater than 1 inch.
(23) The panel **10** further includes a hole **36**. The hole **36** is drilled through the first edge **26** and extends at least partially toward the second edge **30**. The hole **36** is formed by a drilling process as further described below. In the illustrated embodiment, the hole **36** extends from the first edge **26** toward the second edge **30** in a direction perpendicular to the extension direction of the first edge **26**. Stated another way, the hole **36** extends from the first edge **26** toward the second edge **30** along a direction defined by the shortest distance between the first edge **26** and the second edge **30**. In other embodiments, the hole **36** may extend from the first edge **26** toward the second edge **30** along a direction that is not perpendicular to the extension direction of the first edge **26**. For example, the hole **36** may extend diagonally (i.e., extend at an angle with the first edge **26** that is not 90 degrees) from the first edge **26**.
(24) In the illustrated embodiment, the panel **10** includes three holes **36**. In other embodiments, the panel **10** may include fewer or more holes **36**. For example, the panel **10** may include a single hole **36** or may include more than three holes **36**. The illustrated holes **36** all extend from the first edge **26** toward the second edge **30**. As such, the holes **36** are generally parallel to each other. In other embodiments, the holes **36** may extend from different edges (e.g., one hole may extend from the first edge **26** and another hole may extend from the second edge **30**). Additionally or alternatively, the holes **36** may be non-parallel to each other. The holes **36** may have the same length or may have different lengths.
(25) The hole **36** has a hole diameter D.sub.H and a hole length L.sub.H. In some embodiments, the hole diameter D.sub.H may be 5/16 inch (i.e., 0.3125 inches). In other embodiments, the hole diameter D.sub.H may be ½ inch (i.e., 0.5 inches). In still other embodiments, the hole diameter D.sub.H may be between 5/16 and ½ inches. In further embodiments, the hole diameter D.sub.H may be less than 5/16 inch. In even further embodiments, the hole diameter D.sub.H may be greater than ½ inch. The hole length L.sub.H is significantly greater than the hole diameter D.sub.H. In some embodiments, the hole length L.sub.H may be at least 6 inches. In other embodiments, the hole length L.sub.H may be at least 12 inches. In still other embodiments, the hole length L.sub.H may be at least 18 inches. In further embodiments, the hole length L.sub.H may be at least 24 inches. In some embodiments, the hole length L.sub.H may be up to 60 inches. In other embodiments, the hole length L.sub.H may be any value between 6 inches and 60 inches. In further embodiments, the hole length L.sub.H may be greater than 60 inches.
(26) The hole length L.sub.H may be at least 10 times the hole diameter D.sub.H. In some embodiments, the hole length L.sub.H may be at least 12 times the hole diameter D.sub.H. In other embodiments, the hole length L.sub.H may be at least 15 times the hole diameter D.sub.H. In still other embodiments, the hole length L.sub.H may be at least 20 times the hole diameter D.sub.H. In further embodiments, the hole length L.sub.H may be between 10 times and 20 times the hole diameter D.sub.H. In some embodiments, the hole length L.sub.H may at least one fourth of the body length L.sub.B of the body **18** of the panel **10**. In other embodiments, the hole length L.sub.H may be at least one half of the body length L.sub.B. In still other embodiments, the hole length L.sub.H may be at least three-fourths of the body length L.sub.B. In some embodiments, the hole length L.sub.H may be between one fourth and three-fourths the body length L.sub.B. Stated another way, in some embodiments, the hole length L.sub.H may be at least 25% of the body length L.sub.B. In other embodiments, the hole length L.sub.H may be at least 50% of the body length L.sub.B. In still other embodiments, the hole length L.sub.H may be more than 75% of the body length L.sub.B. In further embodiments, the hole length L.sub.H may be up to 90% of the body length L.sub.B. In some embodiments, the hole length L.sub.H may be between 25% and 75% of the body length L.sub.B. In other embodiments, the hole length L.sub.H may be between 25% and 90% of the body length L.sub.B.
(27) In some embodiments, the hole diameter D.sub.H may be at least 30% of the thickness T.sub.B of the body **18** of the panel **10**. In other embodiments, the hole diameter D.sub.H may be at least

50% of the body thickness T.sub.B. In still other embodiments, the hole diameter D.sub.H may be at least 60% of the body thickness T.sub.B. In further embodiments, the hole diameter D.sub.H may be between 30% and 75% of the body thickness T.sub.B.

(28) With reference to FIG. **5**, the hole **36** (FIG. **3**) is configured to receive a structural insert **38** to help support the panel **10** and/or connect the panel **10** to another structure. The illustrated structural insert **38** is a rod assembly. The structural insert **38** includes a rod **42**, a coupling **46**, and a threaded stand **50**. The rod **42** is slidably received in the coupling **46**. The threaded stand **50** is threadedly received in the coupling **46**. In embodiments where the panel **10** includes multiple holes **36**, each hole may receive a structural insert. For example, as shown in FIG. **6**, three structural inserts **38***a*, **38***b*, **38***c* may be received in the panel **10** (FIG. **3**) including three holes **36**. Each of the three holes **36** receives a corresponding one of the three structural inserts **38***a*, **38***b*, **38***c* of FIG. **6**. As illustrated in FIG. **6**, the three structural inserts **38***a*, **38***b*, **38***c* include a middle structural insert **38***a* and two outer structural inserts **38***b*, **38***c*. The middle structural insert **38***a* includes a rod **42***a* that has a longer length than rods **42***b*, **42***c* of the two outer structural inserts **38***b*, **38***c*. In other embodiments, the structural insert **38** may have other configurations for compatibility with the hole **36** of FIG. **3**. For example, the structural insert **38** may have a different rod assembly structure to couple multiple panels **10** together.

(29) Returning reference to FIG. **1**, the workspace **16** includes the screen **14** and a stand **54**. The screen **14** includes five panels **10**, each of which may include one or more of the holes **36** described above. The panels **10** are angled relative to each other. When not in use, the panels **10** may be folded together to reduce the carrying size of the screen **14** for transportation purposes. The stand **54** is placeable on a surface, such as a desk, to prop up the screen **14**. The screen **14**, therefore, functions as a divider for the workspace **16** and may reduce distractions for a worker.

(30) FIG. **7** illustrates a plurality of workspaces **58** according to another embodiment of the disclosure. Many components of the illustrated workspaces **58** are similar to the workspace **16** described above with reference to FIGS. **1-6**. For the sake of brevity, only the differences of the workspaces **16**, **58** are explained below. Each of the workspaces **58** includes a desk **62** and a screen **66**. The screen **66** has two panels **70**. The panels **70** may be similar to the panel **10** described above. The screen **66** surrounds the desk **62** and provides a worker opening **74**. Each of the two panels **70** curves ninety degrees to extend around a corner of the desk **62**. The panels **70** meet at a side of the desk **62** opposite the worker opening **74**. The screen **66** further includes couplings **78** between the panels **70**. Specifically, the screen **66** includes two couplings **78**. The couplings **78** secure the two panels **70** together to surround the desk **62**. The couplings **78** extend between the panels **70** at the point where the panels **70** meet. The panels **70** additionally receive structural inserts **82** that engage a ground surface to support the screen **66** from the ground surface.

(31) FIG. **8** illustrates a workspace **86** according to another embodiment of the disclosure. Many components of the illustrated workspace **86** are similar to the workspaces **16**, **58** described above with reference to FIGS. **1-6** and FIG. **7** respectively. For the sake of brevity, only the differences of the workspaces **16**, **58**, **86** are explained below. The workspace **86** includes a desk **90** and a screen **94**. The desk **90** includes wheeled support stands **98** for improving the case of mobility of the workspace **86**. The wheeled support stands **98** may allow a user to easily push the workspace **86** to a desired location. The screen **94** includes three panels **102***a*, **102***b*, **102***c*. Each panel **102***a*, **102**, **102***c* may be similar to the panel **10** described above. The screen **94** surrounds the desk **90** and provides a worker opening **106**. A first panel **102***a* is positioned opposite the worker opening **106** and curves about the desk **90** at a position adjacent the wheeled support stands **98**. A second panel **102***b* is coupled to the first panel **102***a* at a position adjacent one of the wheeled support stands **98**, and a third panel **102***c* is coupled to the first panel **102***a* at a position adjacent the other of the wheeled support stands **98**. The workspace **86** includes longitudinal couplings **110**. A corresponding coupling **110** extends along each edge of the first panel **102***a* that is touching, or adjacent to, a respective edge of the second panel **102***b* and the third panel **102***c* to couple the

second panel **102***b* and the third panel **102***c* to the first panel **102***a*.
(32) FIG. **9** illustrates a workspace **114** according to another embodiment of the disclosure. Many components of the illustrated workspace **114** are similar to the workspaces **16**, **58**, **86** described above with reference to FIGS. **1-6**, FIG. **7**, and FIG. **8** respectively. For the sake of brevity, only the differences of the workspaces **16**, **58**, **86**, **114** are explained below. The workspace **114** includes a desk **118** and a screen **122**. The desk **118** includes wheeled support stands **126** for improving the ease of mobility of the workspace **114**. The screen **122** includes a panel **130** that curves around the desk **118** to provide a worker opening **134**. The panel **130** may be similar to the panel **10** described above. The panel **130** is foldable at panel flaps **138**. When folded outwards, the panel flaps **138** extend substantially linearly from the panel **130**. The panel flaps **138** additionally extend substantially linearly along the wheeled support stands **126**. When folded together, the screen **122** may improve ease of transportation and therefore allow a user to move the screen **122** to different desks **118**.
(33) FIG. **10** illustrates a drilling kit **142** for drilling into the panel **10**. The drilling kit **142** includes a guide system **146**, a weight **150**, a drill **154**, and a drill bit **158** coupled to the drill **154**. The guide system **146** includes a support rail **162**, a guide block **166**, and a slider **170**. The guide system **146** supports the drill **154** and the drill bit **158** for movement in an axial direction. The axial direction is perpendicular to the first edge **26** of the panel **10**. In the illustrated embodiment, the support rail **162** includes two bars **174**. The bars **174** may also be referred to as elongated bars **174**. The illustrated bars **174** are cylindrical, but may alternatively have other shapes. The bars **174** are spaced apart to fit the drill **154** therebetween. The guide system **146** further includes a stopper **178**. The stopper **178** may be formed of rubber, metal, or another similar material. For example, the stopper **178** may be an O-ring or other type of ring. The stopper **178** is slidably received on one of the bars **174**. The stopper **178** is slidable along the guide system **146** to allow a user to set the maximum distance the drill bit **158** may be inserted into the panel **10**. In the illustrated embodiment, the guide system **146** includes just one stopper **178** slidably received on one of the bars **174**. In some embodiments, the guide system **146** may include two stoppers **178**, each stopper **178** slidably received on one of the bars **174**. In further embodiments, the stopper **178** may have a different configuration or geometric shape for limiting insertion of the drill bit **158** into the panel **10**.
(34) As illustrated in FIG. **11**, the guide block **166** is configured to be positioned adjacent the first edge **26** of the panel **10**. Although the guide block **166** is described as positioned adjacent the first edge **26**, the guide block **166** may be positioned adjacent any edge of the panel **10** into which a user desires to drill. The guide block **166** receives at least a portion of the drill bit **158** to guide movement of the drill bit **158**. Specifically, the guide block **166** is a c-block including a first receiving portion **182** having a first bit receiving aperture **186** and a second receiving portion **190** having a second bit receiving aperture **194**. The user may push the drill bit **158** through the first bit receiving aperture **186** in the first receiving portion **182**, the second bit receiving aperture **194** in the second receiving portion **190**, and into the first edge **26** of the panel **10** to begin drilling the hole **36** (FIG. **2**). The guide block **166** improves and maintains alignment of the drill bit **158** as the drill bit **158** is inserted into the panel **10**. The drilling kit **142** may further include a clamp configured to clamp the guide block **166** on a table **198**, thereby inhibiting movement of the guide block **166** relative to the panel **10**.
(35) Returning reference to FIG. **10**, in the illustrated embodiment, the slider **170** receives the drill **154** and is slidably received on each of the two cylindrical bars **174**. With the slider **170** received on the cylindrical bars **174**, and thus the support rail **162**, and the drill **154** received in the slider **170**, a user may then slide the slider **170** along the support rail **162** to insert the drill bit **158** into the first edge **26** of the panel **10**. The slider **170** is slidable along the support rail **162** toward the first edge **26** of the panel **10** until the slider **170** reaches the stopper **178**, at which point the slider **170** is inhibited from traveling further in the direction of the first edge **26** of the panel **10**.

(36) The weight **150** is a mass of material that can be placed on the panel **10**. In the illustrated embodiment, the weight **150** is a roughly rectangular prism bar. In some embodiments, the weight **150** may be at least 20 pounds. In other embodiments, the weight **150** may be at most 30 pounds. In further embodiments, the weight **150** may be between 20 pounds and 30 pounds. In the illustrated embodiment, the weight **150** is approximately 25 pounds. In other embodiments, the weight **150** may be less than 20 pounds. In further embodiments, the weight **150** may be more than 30 pounds. The weight **150** may be formed of iron, brass, or another similarly dense metal. Alternatively, the weight **150** may be formed of a ceramic material such as porcelain. The panel **10** is oriented parallel to the ground such that a user may place the weight **150** on the first face **20** of the panel **10**. Specifically, a user may place the weight **150** on an area **202** of the first face **20** of the panel **10** that is in axial alignment with the guide system **146**. The weight **150** is configured to apply a force to the area **202** of the first face **20** of the panel **10**. Although the drilling kit **142** of the illustrated embodiment utilizes the weight **150** to apply the force to the area **202** of the panel **10**, the drilling kit **142** may be operable with any other means of applying an adequate force to the area **202** of the panel **10**. For example, the drilling kit **142** may include a hydraulic press that applies a force to the area **202** of the panel **10**. In such examples, the panel **10** may be positioned such that the first face **20** and the second face **22** may extend perpendicular to the ground. Stated another way, the panel **10** may be vertically oriented relative to the ground. As such, the hydraulic press may engage/clamp the panel **10** at each of the first face **20** and the second face **22** to provide the force to the area **202** of the panel **10**. In examples in which a hydraulic press, or any other suitable means for applying a force to the panel **10**, is used in place of the weight **150**, the first face **20** and the second face **22** of the panel **10** may be oriented horizontally, vertically, diagonally, or in any other position relative to the ground.
(37) In the illustrated embodiment, the drill **154** is a conventional drill that may be purchased from an outside supplier of power tools. In other embodiments, the drill **154** may be a dedicated drill specifically designed for the drilling kit **142**. The drill **154** is configured to receive and drive the drill bit **158**. The illustrated drill **154** is battery powered. In other embodiments, the drill **154** may have a cord for connecting to an external power source.
(38) FIGS. **12** and **13** illustrate the drill bit **158**. The drill bit **158** includes a generally hollow body **206** having a smooth outer surface **210**, a drill bit length L.sub.D, a drill bit diameter D.sub.D, a first end **214**, a second end **218**, and an aperture **222**. The aperture **222** extends from the second end **218** and through the generally hollow body **206**. In the illustrated embodiment, the generally hollow body **206** of the drill bit **158** is at most 0.02 inches thick. Specifically, the generally hollow body **206** of the drill bit **158** is 0.013 inches thick. In other embodiments, the generally hollow body **206** of the drill bit **158** is greater than 0.02 inches thick.
(39) The drill bit length L.sub.D extends between the first end **214** and the second end **218**. In some embodiments, the drill bit length L.sub.D is at least 6 inches. In other embodiments, the drill bit length L.sub.D is at least 1 foot. In still other embodiments, the drill bit length L.sub.D is at least 2 feet. In still other embodiments, the drill bit length L.sub.D is at least 3 feet. In further embodiments, the drill bit length L.sub.D may be up to 5 feet. In some embodiments, the drill bit length L.sub.D may be between 6 inches and 5 feet. In some embodiments, the drill bit length L.sub.D may be between 1 foot and 3 feet. In still other embodiments, the drill bit length L.sub.D may be more than 5 feet.
(40) The drill bit diameter D.sub.D is a maximum outer diameter of the generally hollow body **206**, excluding the first end **214** of the drill bit **158**. The drill bit diameter D.sub.D is equal to the diameter D.sub.H of the hole **36** (FIG. **3**) being drilled into the panel **10**. As such, similar to the hole diameter D.sub.H, the drill bit diameter D.sub.D may be 5/16 inches, ½ inch, or between 5/16 inches and ½ inch. In addition, the drill bit diameter D.sub.D may be less than 5/16 inches or may be greater than ½ inch. Similarly, the drill bit length L.sub.D may be at least 10 times the drill bit diameter D.sub.D, at least 20 times the drill bit diameter D.sub.D, or between 10 and 20 times the

drill bit diameter D.sub.D.
(41) The first end **214** of the drill bit **158** is configured to be coupled to the drill **154** (FIG. **10**). Specifically, the drill bit **158** includes a shank **226** at the first end **214** for coupling with the drill **154**. In the illustrated embodiment, the shank **226** is a hex shank. In other embodiments, the shank **226** may be any other type of shank for coupling with the drill **154**. The second end **218** is positioned opposite the first end **214**. The second end **218** has a sharpened edge **230**. Specifically, the sharpened edge **230** is tapered. The taper of the sharpened edge **230** enables the drill bit **158** to cut through polyester during operation of the drill bit **158**. As such, the sharpened edge **230** has a thickness that is less than the thickness of the rest of the generally hollow body **206**. In the illustrated embodiment, the sharpened edge **230** extends approximately 0.75 inches from the second end **218**. The sharpened edge **230** does not include cutting teeth. The aperture **222** extends through the second end **218** of the drill bit **158**. The aperture **22** extends toward the first end **214** of the drill bit **158**, but may end before or at the shank **226**. The aperture **222** may receive material removed by the sharpened edge **230** of the drill bit **158** during drilling, as to be further described herein.
(42) FIG. **14** illustrates a drilling operation **400**. The drilling operation **400** is a method for deep drilling into a panel, such as the panel **10**. Although the drilling operation **400** is described with reference to certain steps, not all of the steps need to be performed or need to be performed in the order presented.
(43) At step **410**, the drilling operation **400** includes providing the guide system **146** (FIG. **10**). The guide system **146** supports the drill **154** having the drill bit **158** for movement in the axial direction. Providing the guide system **146** may include, among other things, providing the support rail **162**, positioning the guide block **166** adjacent the first edge **26** of the panel **10**, and coupling the slider **170** to the support rail **162**.
(44) At step **420** of the drilling operation **400**, a user aligns the guide system **146** with the panel **10**. Specifically, as shown in FIG. **15**, a user may place the guide system **146** on the table **198**, or another similar surface, that supports the panel **10**. In the illustrated embodiment, the guide system **146** includes a front support **234** to be placed on the table **198**. The front support **234** and the guide block **166** of the guide system **146** are placed on the table **198** to position the support rail **162**, and thus the two cylindrical bars **174**, and the guide block **166** in alignment with the panel **10**. Further, the user may place the first bit receiving aperture **186** and the second bit receiving aperture **194** into alignment with a desired hole location for the panel **10**. In some embodiments, the user may additionally place a clamp such as a c-clamp in engagement with the guide block **166** and the table **198** to inhibit the guide block **166** from moving relative to the panel **10**.
(45) At step **430** of the drilling operation **400**, a force is applied to the face **20** of the panel **10**. Specifically, as shown in FIG. **16**, a user may place the weight **150** on the panel **10** at the area **202** corresponding to the desired hole location to apply the force to the face **20** of the panel **10**. The user specifically places the weight **150** at the area **202** on the first face **20** of the panel **10** that is in axial alignment with the guide system **146**. Although the method for deep drilling into the panel **10** of the illustrated embodiment discloses placing the weight **150** on the panel **10** after the guide system **146** has been provided and aligned with the panel **10**, a user may alternatively place the weight **150** on the panel **10** before the guide system **146** has been provided and aligned with panel **10**. As such, the user may provide and align the guide system **146** with the area **202** that is in axial alignment with the weight **150** in such instances.
(46) The weight **150** provides a force to the area **202** on the first face **20** of the panel **10** that compresses the polyester fibers of the panel **10** to assist in maintaining a constant direction of movement through the body **18** of the panel **10** for the drill bit **158** during drilling. Compressing the panel **10** inhibits the drill bit **158** from traveling away from an intended path during drilling. In absence of the weight **150**, the fibers are not compressed, which provides relatively less resistance for the drill bit **158** such that the drill bit **158** may wobble and/or travel off course during drilling.

As previously disclosed, the drilling kit **142** (FIG. **10**) may be operable with any force providing means that is capable of compressing the polyester fibers of the panel **10**.

(47) With reference to FIGS. **16** and **17**, prior to step **440**, a user may cut a slit into the first face **20** of the panel **10**. In alternative embodiments, the slit may be cut into the second face **22** (FIG. **4**) of the panel **10**. A user may use a paring knife or any other similar cutting tool capable of making a precise and accurate cut to create the slit. In some embodiments, the slit may be at least 0.7 inches long. In other embodiments, the slit may be at most 0.7 inches long. In further embodiments, the slit may have a length that is dependent on the drill bit diameter D.sub.D (FIG. **13**). The slit may be cut by the user to extend into the body **18** from the first face **20**. In some embodiments, the slit may extend at least 60% into the body **18** from the first face **20**. In other embodiments, the slit may extend at least 70% into the body **18** from the first face **20**. In further embodiments, the slit may extend at least 80% into the body **18** from the first face **20**. In even further embodiments, the slit may extend at least 90% into the body **18** from the first face **20**. With additional reference to FIG. **3**, the position of the slit may be dependent on an intended hole length L.sub.H. For example, if a user intends to drill a hole **36** having a hole length L.sub.H extending 30 inches from the first edge **26** of the panel **10**, then the slit may be created, that is, cut, 30 inches from the first edge **26** of the panel **10** in axial alignment with the intended hole **36**. In another example, if a user intends to drill a hole **36** having a hole length L.sub.H extending 40 inches from the first edge **26** of the panel **10**, then the slit may be created, that is, cut, 40 inches from the first edge **26** of the panel **10** in axial alignment with the intended hole **36**.

(48) At step **440** of the drilling operation, the hole **36** (FIG. **3**) is drilled into the panel **10**. Specifically, as shown in FIGS. **17** and **18**, the hole **36** is drilled using the guide system **146** and the drill **154** while the force is applied to the face **20** of the panel **10**. To drill the hole **36**, a user may push the slider **170** toward the first edge **26** to insert the drill bit **158** into the first edge **26**. In some embodiments, the drill bit **158** may be inserted into the first edge **26** of the panel **10** at a rate of approximately 1 inch per second. In some embodiments, the optimal rate of insertion for the drill bit **158** may be less than 1 inch per second. For example, the drill bit **158** may be inserted into the first edge **26** of the panel **10** at a rate of ½ inch per second. In other embodiments, the optimal rate of insertion may be more than 1 inch per second. For example, the drill bit **158** may be inserted into the first edge **26** of the panel **10** at a rate up to 2 inches per second.

(49) Drilling the hole **36** of FIG. **3** further includes rotating the drill bit **158** with the drill **154**. In some embodiments, the drill bit **158** may be rotated at a rate of at least 800 rpm. In other embodiments, the drill bit **158** may be rotated at a rate of at most 1100 rpm. In still other embodiments, the drill bit **158** may be rotated at a rate between 800 rpm and 1100 rpm. In the illustrated embodiment, the drill bit **158** may be rotated at a rate of approximately 950 rpm. In other embodiments, the drill bit **158** may be rotated at a rate that is less than 800 rpm. In further embodiments, the drill bit **158** may be rotated at a rate greater than 1100 rpm.

(50) A user may push the slider **170** (and thereby the drill **154**) toward the first edge **26** of the panel **10** until the slider **170** reaches the stopper **178**. FIG. **18** illustrates the slider **170** as the slider **170** reaches the stopper **178**. The stopper **178** inhibits the user from the pushing the slider **170** any further toward the first edge **26** of the panel **10**. In other embodiments, the guide system **146** may not include the stopper **178**.

(51) Once the drill bit **158** has been inserted a desired distance into the panel **10**, the user may pull the drill bit **158** from the panel **10** to remove the drill bit **158** from the panel **10**. Once the drill bit **158** has been entirely removed and the drill **154** has been turned off, the user may then remove the slider **170** from the support rail **162** and, additionally or alternatively, remove the drill bit **158** from the slider **170**. The user may then remove material from the aperture **222** (FIG. **13**) of the drill bit **158**. Specifically, with reference to FIG. **13**, the drill bit **158** receives, in the aperture **222**, polyester fibers drilled (i.e., removed) by the sharpened edge **230** during the drilling operation **400**. The slit that is cut into the one of the first face **20** and the second face **22** enables retention of material in

the drill bit **158**. In the absence of the slit, the material cut by the drill bit **158** would remain connected to the rest of the panel **10** along the axial direction. The slit therefore provides an end of the hole **36** (FIG. **3**) in which the material removed during the drilling process is severed from the rest of the panel **10**. As such, the slit enables the material to be adequately retained in the drill bit **158** and thus, adequately removed from the panel **10**. Returning reference to FIG. **18**, with the material removed, the user may move the guide system **146** to a new desired hole location and repeat the drilling operation **400** (FIG. **14**), as described here above, to drill as many holes **36** (FIG. **3**) as desired in the panel **10**.
(52) In some embodiments, multiple holes **36** may be drilled into the panel **10** simultaneously. In such embodiments, multiple drilling kits **142** may be provided to drill the multiple holes **36** (FIG. **3**) simultaneously. Alternatively, the support rail **162** and/or the slider **170** may be modified (e.g., elongated) to support multiple drills, which can then drill multiple holes simultaneously.
(53) Although the invention has been described with reference to certain embodiments, variations and modifications exist within the spirit and scope of the invention. Various features of the invention are set forth in the following claims.

Claims

1. A method for deep drilling into a panel composed of compressed polyester fibers, the panel having an edge and a face, the method comprising: providing a guide system, the guide system supporting a drill having a drill bit for movement in an axial direction; aligning the guide system with the panel; placing a weight on the face of the panel, the weight being placed on an area of the panel that is in axial alignment with the guide system and spaced apart from the edge of the panel; and drilling a hole into the edge of the panel with the guide system and the drill while the force is applied to the face of the panel, wherein the edge of the panel is a first edge, wherein the panel has a second edge opposite the first edge and a length measured between the first edge and the second edge, and wherein placing the weight includes positioning the weight to extend a majority of the length of the panel in axial alignment with the guide system.
2. The method of claim 1, wherein the drill bit includes a hollow body having a smooth outer surface and a sharpened edge without cutting teeth, and wherein drilling the hole includes receiving part of the panel in the generally hollow body.
3. The method of claim 1, wherein drilling the hole includes inserting the drill bit into the edge of the panel at a rate of approximately 1 inch per second.
4. The method of claim 1, wherein drilling the hole includes rotating the drill bit with the drill at a rate of 800 to 1100 rpms.
5. The method of claim 1, wherein drilling the hole into the edge of the panel includes drilling a hole having a length of at least 6 inches.
6. The method of claim 5, wherein the length of the hole is at least 10 times a diameter of the hole.
7. The method of claim 5, wherein the edge of the panel is a first edge, wherein the panel has a second edge opposite the first edge and a length measured between the first edge and the second edge, and wherein the length of the hole is at least half the length of the panel.
8. The method of claim 1, wherein the panel has a thickness between 0.5 inches and 1 inch.
9. The method of claim 1, wherein providing the guide system includes providing a support rail, positioning a guide block adjacent the edge of the panel, the guide block receiving at least a portion of the drill bit to guide movement of the drill bit relative to the panel, and coupling a slider to the support rail, the slider supporting the drill.
10. The method of claim 1, wherein drilling the hole includes moving the slider along the support rail in the axial direction toward the panel.
11. The method of claim 1, wherein the weight is at least 20 pounds.